

QR Code Based Personal Information Sharing Application

Master's Dissertation Progress Report

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Chapter 1

Introduction and Motivation

The increasing need for secure digital identity exchange and privacy-preserving information sharing has created demand for solutions that minimize reliance on centralized service providers. Traditional approaches often require users to trust third-party platforms with sensitive personal or business information, introducing privacy, security, and availability concerns.

This project proposes a decentralized system for sharing personal and business information using QR codes, asymmetric cryptography, and direct peer-to-peer communication. The objective is to enable users to exchange sensitive information securely while maintaining full control over their data. The system supports both small data transfers embedded directly into QR codes and larger transfers performed through encrypted peer-to-peer channels.

The proposed approach combines ease of use with strong security guarantees. QR codes provide a familiar user interaction model, while asymmetric encryption ensures that shared information can only be accessed by the intended recipient.

The following chapter introduces the data format used by the system and the types of information that can be exchanged.

Chapter 2

Data Format

The data being transferred is structured data, similar to the information encoded by a vCard, and is compatible with a form-based format. The data is organized as key-value pairs using the JSON format.

The system supports sharing encrypted personal information using asymmetric cryptography. The shared information can include contact information similar to a vCard, additional sensitive personal information, company information, and larger files such as personal or business documents.

2.1 Example of Shared Business Information

```
{  
  "company": "Acme Solutions",  
  "owner": "John Smith",  
  "phone": "+1-555-123-4567",  
  "email": "john@acmesolutions.com",  
  "website": "https://acmesolutions.com",  
  "address": "123 Main Street, New York, NY"  
}
```

2.2 Examples of Larger Files

- Digital scans of personal or business documents
- Contracts and NDAs
- Financial information

When the shared information is only a few kilobytes in size, it can be encrypted, embedded directly into a QR code, and shared with the intended receiver. If the data is too large to fit into a QR code, or if a secure peer-to-peer connection is preferred, the data is transferred through an encrypted peer-to-peer channel.

Chapter 3

System Architecture and Design

The proposed architecture consists of a sender application and a receiver application operating in a peer-to-peer communication model. Information can be transferred directly without intermediary entities using one of two methods:

1. Direct data exchange through secured QR codes.
2. Data exchange through a secured peer-to-peer connection, where QR codes are used only to establish the secure connection.

Data storage is decentralized and remains exclusively on the sender's device. Information is shared only after the sender reviews and approves the requested data.

3.1 Direct Data Exchange Through Secured QR Codes

3.1.1 Data Transfer Protocol

Step 0 (Optional): Sender offers to share arbitrary data with a receiver.

Step 1: The receiver requests specific information and generates a temporary asymmetric key pair. The public key is embedded in the QR code.

Step 2: The sender scans the receiver's QR code, reviews the request, selects the information to share, encrypts it with the receiver's public key, and generates a QR code containing the encrypted payload.

Step 3: The receiver scans the sender's QR code, decrypts the information using the matching private key, stores the data locally, and securely deletes the temporary key pair.

3.1.2 Managing Multiple Transfers

A transfer session identifier is embedded in the protocol messages to allow both applications to manage multiple concurrent transfer sessions.

3.2 Data Transfer Through a Secured Peer-to-Peer Connection

This method is intended for larger data transfers or situations where embedding the data directly inside the QR code is not practical.

3.2.1 Data Transfer Protocol

Step 0 (Optional): Sender offers to share arbitrary data.

Step 1: Receiver requests specific information.

Step 2: The sender prepares the encrypted package and generates a QR code containing a temporary URI and the parameters required to establish a secure connection.

Step 3: The receiver scans the QR code, decrypts the connection information, connects directly to the sender's device, downloads the encrypted package, decrypts it locally, and securely deletes the temporary key pair.

An optional authentication mechanism can be used where the receiver digitally signs a challenge using its private key. The sender verifies the signature using the receiver's public key before allowing the download.

Chapter 4

Security Design

The security design will focus on:

- End-to-end encryption using asymmetric cryptography.
- Temporary cryptographic key generation for each transfer session.
- Decentralized data storage.
- Mutual verification of transfer participants.
- Protection against unauthorized data disclosure.
- Secure deletion of temporary cryptographic material.

Chapter 5

Conclusion

The proposed system combines QR-code-based interactions with modern cryptographic techniques to enable secure, decentralized sharing of personal and business information. By supporting both direct QR transfers and encrypted peer-to-peer communication, the system can accommodate a wide range of information-sharing scenarios while preserving user privacy and data ownership.

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